

METABOLIC RESPONSES TO A BATTLING ROPE PROTOCOL PERFORMED IN THE SEATED OR STANCE POSITIONS

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ABSTRACT

Brewer, W, Kovacs, R, Hogan, K, Felder, D, and Mitchell, H. Metabolic responses to a battling rope protocol performed in the seated or stance positions. *J Strength Cond Res* 32(12): 3319–3325, 2018—The purpose of this study is to compare the levels of oxygen consumption and heart rate responses elicited by a treadmill or cycle $\dot{V}O_{2\max}$ test with a standing or sitting battling rope protocol (BRP) (treadmill vs. standing BRP) (cycle vs. seated BRP). Forty healthy subjects performed either a ramped $\dot{V}O_{2\max}$ treadmill or cycle test. At least 3 days later, the subjects who performed the ramped treadmill test did the standing BRP, and the subjects who performed the ramped cycle test did the sitting BRP. Each BRP consisted of 10 sets of 15/45 seconds of work/rest for 10 sets. $\dot{V}O_{2\text{peak}}$ and HRpeak were recorded. Metabolic responses were significantly lower for the sitting BRP and standing as compared to the HRmax and $\dot{V}O_{2\max}$ values derived from the bicycle and treadmill tests. The BRP produced a $\dot{V}O_{2\text{peak}}$ that was 71.87% (sitting) and 68.37% (standing) of the subjects' $\dot{V}O_{2\max}$ assessed on the bicycle and treadmill protocol. Moderate correlations were found between the $\dot{V}O_2$ during the seated ($r = 0.61$; $p = 0.003$) and standing ($r = 0.43$; $p = 0.03$) BRP and the bicycle and treadmill $\dot{V}O_{2\max}$ tests, respectively. The HRpeak elicited by the BRP performed in sitting ($r = 0.52$; $p = 0.009$) and standing ($r = 0.67$; $p = 0.001$) had a moderate correlation with the HRmax derived from the bicycle and treadmill tests. Battling ropes may be a low cost, accessible option to improve cardiovascular endurance for individuals who cannot stand or move their lower extremities in a rhythmic manner to conduct aerobic exercise.

KEY WORDS high-intensity, interval training, cardiovascular exercise

INTRODUCTION

Obtaining sufficient amounts of physical activity is a challenge for many individuals who live in industrialized societies. The federal guidelines suggest that all individuals obtain 150 minutes of moderate to vigorous physical activity that stresses the cardiovascular system each week (13). Individuals who are unable to stand, walk, run, or move their lower extremities in a rhythmic manner due to balance deficits, joint diseases, or neuromuscular weakness have greater difficulty achieving these guidelines. According to the Healthy People 2010 report, 56% of adults with disabilities do not engage in any leisure-time physical activity compared to 36% among adults without disability (4). There are a multitude of barriers faced by individuals who are disabled toward the attainment of health-enhancing physical activity (19). The muscles of the lower extremities by dry mass represent the largest in the body. Diseases that affect the neurological or musculoskeletal systems, which results in impairment of muscular force, joint motion, neuromuscular coordination, and balance, have adverse effects on the ability to participate in aerobic activities.

The employment of the upper extremities has been used extensively in rehabilitation programs for individuals with lower-extremity impairments. Previous studies have shown that the use of equipment such as upper-extremity ergometry results in increases in peak oxygen consumption, heart rate (HR), and ratings of perceived exertion (RPEs), which would increase cardiovascular fitness (9,14,22,25). These studies noted that upper-body ergometry typically produces increases in oxygen consumption and anaerobic threshold that is approximately 34% less than lower-extremity aerobic exercises. The use of the upper-body ergometer to maintain or improve cardiovascular fitness for individuals who are disabled reveals this modality as promising. The costs of these machines and the potential for mechanical wear and tear must be considered because these 2 factors may prohibit the use of upper-body ergometry for certain individuals.

Battling ropes exercise is a tool that has been used by strength and conditioning specialists to promote improvements in general cardiovascular fitness and local muscular endurance of the upper extremities. Battling ropes are

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a low-cost alternative to other exercise modalities with purchase prices ranging from \$16 to \$50. Published reports have documented that the use of battling ropes is considered a vigorous-intensity exercise that promotes increases in HR that can peak within 86% of age-matched norms (7,16,17). It has also been shown that battling ropes can produce the large, acute metabolic responses when compared to free-weight or body weight exercises. These acute changes in oxygen consumption, HR, ventilation, and perceptions of exertion studied were conducted in weight-bearing positions. Weight-bearing positions such as standing and squatting typically evoke larger responses than nonweight-bearing positions such as sitting.

There are numerous protocols that vary in work-to-rest ratios and types of swings used for battling rope exercise. These protocols are based on the principles of high-intensity interval training (HIIT), which involves alternating periods of highly intense ($>60\%$ of $\dot{V}O_{2\max}$) work bouts that may last from 10 seconds to 4 minutes in length with low-to-moderate intensity rest periods ($40\text{--}60\%$ of $\dot{V}O_{2\max}$) that typically ranges from 1 to 2 times the length of the work bout. Several studies have demonstrated the effectiveness and safety of HIIT in various clinical populations that range from individuals with cancer, heart failure, postmyocardial infarction, coronary artery disease, percutaneous coronary interventions, and coronary bypass grafting (1,2,5,8,24). Battling rope protocols (BRPs) that use work-to-rest ratios, as previously described, have the potential to improve cardiovascular fitness; however, to date, the metabolic changes produced by them have not been compared to traditional forms of cardiovascular exercise such as the bicycle or treadmill. The metabolic demands on the cardiovascular system are influenced by whether an individual is in a weight-bearing position or not. Understanding how the metabolic demands are altered during a BRP when individuals assume the seated or standing position could elucidate how this low-cost alternative for cardiovascular exercise could be beneficial for a clinical population that often has difficulty assuming weight-bearing positions.

The metabolic perturbations produced by battling ropes for individuals who are unable to stand have not been studied. If these acute metabolic changes induce cardiovascular effects for individuals who are unable to stand, this would provide another tool for the strength and conditioning professional to use to promote fitness for the multitude of individuals with this dysfunction. The purpose of this study is to compare the levels of oxygen consumption, HR responses, and RPE elicited by a $\dot{V}O_{2\max}$ test performed either on a treadmill or bicycle with a rope protocol performed while in the standing or seated position, respectively.

Hypotheses: (a) The battling ropes will elicit statistically similar responses in oxygen consumption, HR elevations, and RPE as compared to respective $\dot{V}O_{2\max}$ test performed in a similar position as the rope protocol. (b) There will not be a significant difference in oxygen consumption between

the sitting and standing BRPs. (c) Battling ropes will prove to be a viable option for cardiovascular exercise for individuals who cannot ambulate or ride a bike due to conditions such as poor balance, weakness, joint dysfunctions, or spinal cord injury.

METHODS

Experimental Approach to the Problem

Each subject was randomly allocated to either the bicycle $\dot{V}O_{2\max}$ test and the seated BRP or the treadmill $\dot{V}O_{2\max}$ test and the standing BRP. The BRP were conducted on a separate day from the $\dot{V}O_{2\max}$ tests. Heart rate, oxygen consumption, and RPE were measured throughout the $\dot{V}O_{2\max}$ tests and after each round of the BRP.

Subjects

Subjects consisted of healthy, young adults recruited from Texas Woman's University-Houston campus. Inclusion criteria: age between 18 and 32 years. A subject was excluded if he/she: (a) had current complaints of lower- or upper-extremity pain or musculoskeletal disorder that precluded the walking or running on a treadmill, riding a stationary bicycle, or grasping and swinging the weighted ropes against the floor; (b) was diagnosed with a chronic or acute cardiovascular, metabolic, renal, or pulmonary disease; and (c) answered "yes" to one or more questions posed on the Modified PAR-Q questionnaire (21). This study was approved by the Texas Woman's University Institutional Review Board. Each subject was informed of the benefits and risks of this investigation by signing the institutionally approved informed consent document before any data collection procedures used to participate in this study.

Procedures

Before testing, each subject signed an informed consent and completed the Modified Physical Activity Readiness Questionnaire (PAR-Q) to ensure each subject was safe to participate in the study. Each subject was randomly assigned to conduct either a maximal walk/run test on the treadmill and perform a standing BRP or a lower-body cycle ergometer test and perform a seated BRP. Protocol assignment was based on a pregenerated computer randomization table and there was at least a 3-day time interval between the maximal exercise tests on the treadmill or bicycle and the respective BRP.

During the first session, each subject's blood pressure, HR, height, and body mass measurements were taken and recorded. Height and body mass were measured using a standard stadiometer and calibrated scale, respectively. Subjects were fitted with a Hans Rudolph reusable $\dot{V}O_2$ facemask $\dot{V}O_2$ and a Polar HR monitor so that vitals could be recorded throughout the exercise tests.

A ramped Bruce Protocol $\dot{V}O_{2\max}$ was used for the treadmill test. This test consisted of 46 stages with the inclination or the speed increasing every 20 seconds until the subject was unable to maintain the walking pace. Every

3 minutes, the subject's blood pressure and RPE were recorded to monitor exercise intensity and the subjects' safety. The test was terminated if the subject requested to stop, a respiratory exchange ratio of >1.0 was achieved, or if the subject displayed adverse cardiovascular signs or symptoms.

A ramped maximal bicycle test using the YMCA protocol with the Monark 828E lower-body ergometer was used for the seated $\dot{V}O_2$ max test. This test consisted of a 3-minute warm-up at a rate of 50 revolutions per minute with a workload of 50 watts. Subjects were to maintain a cadence of $100 \text{ b} \cdot \text{min}^{-1}$ (bpm) with the use of a metronome. After the warm-up, the workload increased by 25 W each minute until the subject was unable to maintain the pedal rate/cadence, the respiratory exchange ratio of >1.0 was achieved, or if signs and symptoms that were indicative of an adverse health event were present. Every 3 minutes, the patient's blood pressure and RPE were taken.

Battling Rope Protocol. The BRP was performed during the second session at least 3 days after the initial session. The ropes used were 50-foot long and 1.5 inch in diameter. For the treadmill/standing BRP group, subjects were positioned with the knees and hips flexed to approximately 45° and 60° , respectively, with one end of the rope held in each hand (Figure 1).

For the bicycle/seated BRP group, subjects positioned themselves in a chair with hips flexed to approximately 120° , lumbar spine in neutral position, knees in 90° degrees of flexion, and feet flat on the floor at shoulder width apart (Figure 2).



Figure 1. Standing battling rope position.



Figure 2. Seated battling rope position.

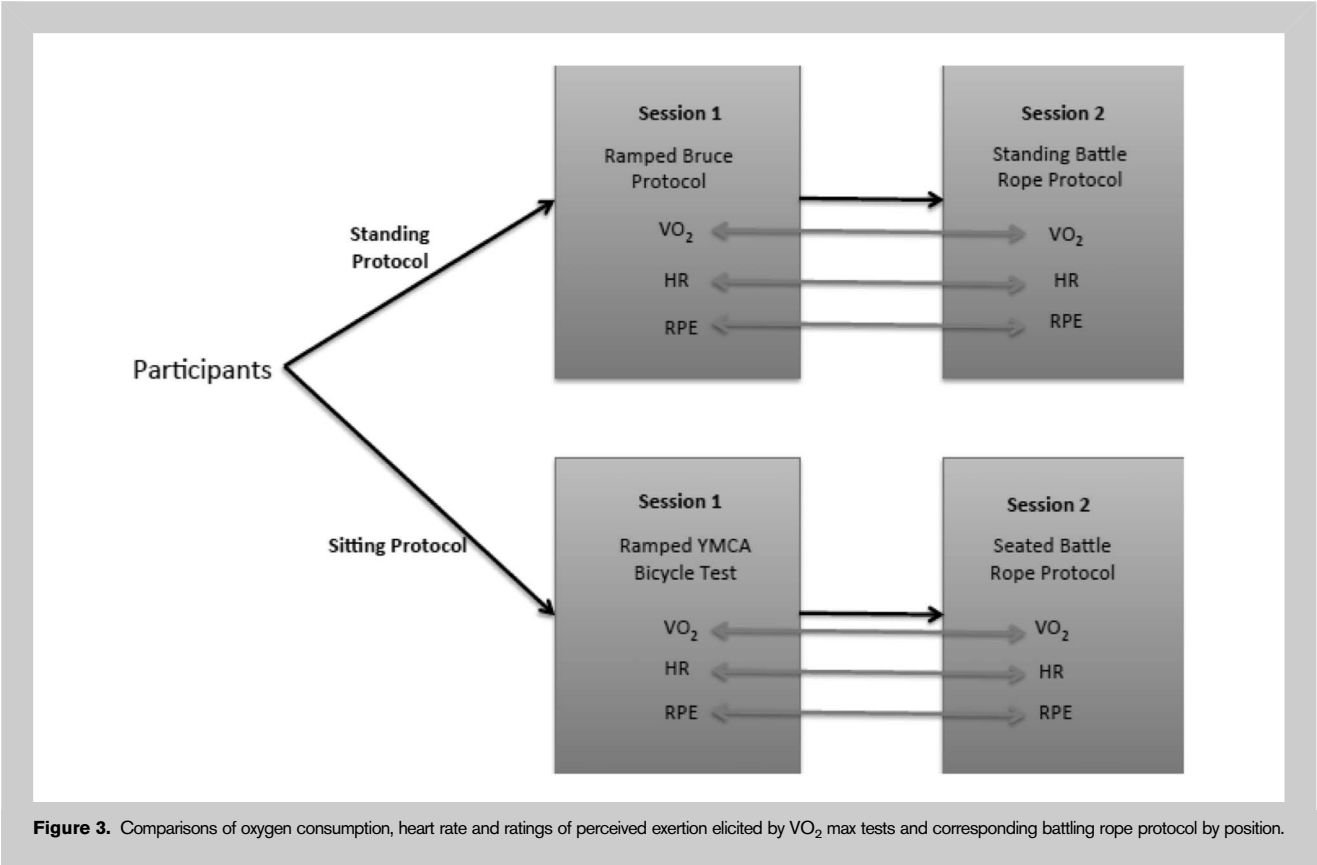
Each subject was instructed to perform double-arm swings at set $100 \text{ b} \cdot \text{min}^{-1}$ cadence for 15 seconds for each work bout, followed by 45 seconds of rest. Before commencement, a simulated 3-minute warm-up period was given to allow each subject to practice and adjust as needed. Each protocol consisted of 10 cycles of 15 seconds of work followed by 45 seconds of rest for a total time of 10 minutes. At the end of each cycle, the RPE, HR, $\dot{V}O_2$, and BP were retrieved from each subject. The testing was stopped if the subject could not complete the rope swing protocol at the required cadence, the subject requested to stop, or if the subject demonstrated signs/symptoms that could indicate an adverse health event.

Outcome Measures. $\dot{V}O_2$ max and HR were assessed using a metabolic cart and HR monitor, respectively, during the treadmill or bicycle $\dot{V}O_2$ max tests and the respective BRP. All subjects were able to achieve a respiratory exchange ratio >1.0 signifying an achievement of $\dot{V}O_2$ max. Rating of perceived exertion was obtained verbally after each minute of exercise.

Statistical Analyses

Figure 3 displays the comparisons that will be made between the maximal exercise test and the respective BRP.

Mean comparisons of the peak RPE, $\dot{V}O_2$, and HR between the seated and standing BRP with those obtained from the bicycle and treadmill $\dot{V}O_2$ max tests were, respectively, conducted using 2 tailed, paired t tests. Bivariate correlation coefficients were used to determine the strength of the associations between peak RPE, $\dot{V}O_2$, and HR obtained from the seated and standing BRPs with those from the bike and treadmill $\dot{V}O_2$ max tests,



respectively. One-tailed, Pearson product-moment correlation coefficients were calculated for variables that met the assumptions of normality. For variables that departed from this assumption, 1-tailed Spearman's rho correlation coefficients were calculated. Level of significance was set at 0.05 for all statistical analyses. Data were analyzed using IBM SPSS Version 24.

RESULTS

Forty subjects (24 females/16 males) with a mean (SD) age of 24.8 (2.4) years participated in the study. The subjects' anthropometric data are presented in Table 1.

Twenty subjects (10 women/10 men) were randomly allocated to have their metabolic responses to the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test compared to the seated BRP (Table 2). A significant difference was found between the levels of oxygen consumption ($\dot{\text{V}}\text{O}_2$) induced by the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test and the seated BRP ($p < 0.001$). The oxygen consumption was approximately $11 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ lower for the seated BRP as compared to the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test. A moderate correlation was found between the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test and the seated BRP ($r = 0.61$; $p = 0.003$). The increase in HR induced from the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test was significantly higher than the peak HR derived from the seated BRP ($p = 0.001$). The HR elicited from the seated BRP was approximately $15 \text{ b} \cdot \text{min}^{-1}$ lower than that elicited from the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test. A moderate correlation was found between the HR produced by the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test and that of the seated BRP ($r = 0.43$; $p = 0.03$). A nonsignificant mean difference was found between the RPE generated from the bicycle $\dot{\text{V}}\text{O}_{2\text{max}}$ test and the seated

TABLE 1. Height, body mass, age, % body fat, and lean body mass.

	Men $n = 16$, mean (SD)	Women $n = 24$, mean (SD)
Height (cm)	175.58 (6.58)	164.66 (8.23)
Body mass (kg)	81.76 (11.18)	66.28 (9.71)
Age (y)	25.25 (2.65)	24.54 (2.26)
Body fat %	18.14 (7.94)	26.42 (5.80)
Lean body mass (kg)	66.35 (6.75)	50.83 (9.18)

TABLE 2. Comparisons of the levels of oxygen consumption ($\dot{V}O_2$), HR, and RPE of the bicycle (BK) $\dot{V}O_{2\max}$ test and the seated BRP.*

	Mean (SD)	<i>t</i> value (df), <i>p</i>	<i>r</i> , <i>p</i>
Resting HR	64.4 (11.45)		
Average HR (seated BRP)	146.52 (16.37)		
Average $\dot{V}O_2$ (seated BRP)	18.84 (5.74)		
$\dot{V}O_{2\text{peak}}$ (seated BRP)	24.23 (7.28)	5.31 (18), <0.001	0.61, 0.003
$\dot{V}O_{2\max}$ (BK $\dot{V}O_{2\max}$ test)	35.99 (11.62)		
% of $\dot{V}O_{2\max}$	67.3%		
HRmax (BK $\dot{V}O_{2\max}$ test)	178.5 (11.39)		
HRpeak (seat BRP)	163.9 (15.73)	3.78 (18), 0.001	0.43, 0.03
HRpeak/HRmax%	91.8%		
Peak RPE (BK $\dot{V}O_{2\max}$ test)	18.25 (1.25)		
Peak RPE (seat BRP)	17.74 (1.52)	−0.936 (19), 0.361	0.54, 0.008

*BRP = battling rope protocol; HR = heart rate; RPE = rating of perceived exertion.

BRP. A moderate correlation was found between the bicycle $\dot{V}O_{2\max}$ test and the seated BRP ($r = 0.54$; $p = 0.008$) for RPE. The peak HR from the seated BRP was approximately 72% of the maximum HR produced by the bicycle $\dot{V}O_{2\max}$ test.

The remaining 20 subjects (14 females/6 males) were randomly allocated to the treadmill $\dot{V}O_{2\max}$ test compared to the standing BRP (Table 3). A significant difference was found between the levels of oxygen consumption induced by the treadmill $\dot{V}O_{2\max}$ test and the standing BRP ($p < 0.001$). The oxygen consumption was approximately 14 ml·kg^{−1}·min^{−1} lower for the standing BRP as compared to the treadmill $\dot{V}O_{2\max}$ test. A moderate correlation was found between the treadmill $\dot{V}O_{2\max}$ test and the standing

BRP ($r = 0.52$; $p = 0.009$) for $\dot{V}O_2$. The increase in HR induced from the treadmill $\dot{V}O_{2\max}$ test was significantly higher than the peak HR derived from the standing BRP ($p < 0.001$). The HR elicited from the standing BRP was approximately 13 b·min^{−1} lower than that elicited from the treadmill $\dot{V}O_{2\max}$ test. A moderate correlation was found between the HR produced by the treadmill $\dot{V}O_{2\max}$ test and the standing BRP ($r = 0.67$; $p = 0.001$). A nonsignificant mean difference and weak correlation was found between the RPE generated from the treadmill $\dot{V}O_{2\max}$ test and the standing BRP. The peak HR from the standing BRP was approximately 68% of the maximum HR produced by the treadmill $\dot{V}O_{2\max}$ test.

TABLE 3. Comparisons of the levels of oxygen consumption ($\dot{V}O_2$), HR, and RPE of the treadmill (TM) $\dot{V}O_{2\max}$ test and the standing BRP.*

	Mean (SD)	<i>t</i> value (df), <i>p</i>	<i>r</i> , <i>p</i>
Resting HR	64.7 (10.80)		
Average HR (stand BRP)	157.265 (14.14)		
Average $\dot{V}O_2$ (stand BRP)	21.22 (5.95)		
$\dot{V}O_{2\text{peak}}$ (stand BRP)	26.56 (6.18)	6.19 (19), <0.001	0.52, 0.009
$\dot{V}O_{2\max}$ (TM $\dot{V}O_{2\max}$ test)	40.85 (11.32)		
% of $\dot{V}O_{2\max}$	65.02%		
HRmax (TM $\dot{V}O_{2\max}$ test)	184.75 (9.27)		
HRpeak (stand BRP)	171.1 (13.79)	6.37 (19), <0.001	0.67, 0.001
HRpeak/HRmax%	92.6%		
Peak RPE (TM $\dot{V}O_{2\max}$ test)	18.05 (1.43)		
Peak RPE (stand BRP)	17.3 (2.8)	1.44 (19), 0.167	0.21, 0.182

*BRP = battling rope protocol; HR = heart rate; RPE = rating of perceived exertion.

DISCUSSION

This study compared the RPE, $\dot{V}O_2$, and HR responses elicited by a treadmill or bike $\dot{V}O_{2\max}$ protocol as compared to a BRP in standing and sitting, respectively. Our main findings were that the metabolic responses were significantly lower for both BRPs as compared to the bicycle and treadmill tests. Previous studies have shown that aerobic exercise with use of the upper extremities have a lower metabolic demand but higher levels of perceived exertion as compared to aerobic exercise with the lower extremities (3,15). The upper extremities tend to have a much lower quantity of muscle mass than the lower extremities, reducing the demands on the cardiovascular system. This study found that the perceptions of exertion were lower for both the seated and standing BRPs as compared to their respective $\dot{V}O_{2\max}$ tests. This may be reflective of the interval-type program that was used for the BRP as compared to the continuous, progressive nature of the $\dot{V}O_{2\max}$ tests.

The peak HRs achieved as a percentage of the maximum HR were not significantly different between the standing and sitting BRP. This was an unexpected finding, considering the stance position increases the weight-bearing demands, muscular requirements to stabilize the trunk, and the ability to use the trunk and lower extremities to provide momentum during the rope swings. The standing and sitting BRPs produced mean peak HRs as a percentage of the maximum that were approximately 92 and 93%, respectively, whereas the mean peak $\dot{V}O_2$ as a percentage of the maximum were approximately 65 and 67% for the standing and seated battling ropes, respectively. These intensities would be recommended for most individuals to use when engaging in aerobic exercise on a frequent basis to promote health using vigorous intensities (18). A similar finding was documented in a study by Faigenbaum et al., who conducted a study that examined the peak HR and $\dot{V}O_2$ as a percentage of their maximum for boys with a mean of age of 10.6 years who performed a 10-minute BRP that used 5 different swing techniques. They found peak HRs as a percentage of the maximum to range between 52.9 and 86.4% and 21.5 and 64.8% for the percentage of $\dot{V}O_{2\max}$ (6).

Heart rate variability is a desirable cardiac autonomic response that is reflective of parasympathetic activity during resting states and has been shown in numerous studies to correlate with cardiovascular fitness (11,23). A recent systematic review that examined HR variability during and after bouts of exercise noted that cardiovascular exercise conducted by the upper extremities had a more rapid withdrawal of parasympathetic activity than exercise conducted with the lower extremities (12). Although the HRs induced by the BRPs were significantly lower than their respective maximal exercise tests performed on the bicycle and treadmill, the elevated HRs seen with the BRPs may be due to a more rapid parasympathetic withdrawal. Electrocardiography was not performed in this study, which limits the

mechanistic understanding of the impact of this cardiac autonomic response on the HRs observed during this study.

Both the sitting and standing BRPs could offer an alternative form of exercise for individuals who cannot perform rhythmic movements with their legs. Other studies have found lasting benefits from upper-extremity interval training. In fact, one study found that regular interval training with a seated double-poling ergometer (SDPE) increased oxygen uptake and power output in participants with a spinal cord injury below T5 (10).

In another study, sprint intervals on the SDPE showed greater improvements in $\dot{V}O_{2\max}$ in cross-country skiers with the diagonal technique and upper-body maximal strength than continuous endurance training (26). In addition, another study found that handcycling performance improved significantly in upper-body endurance capacity with HIIT than moderate intensity or the control group (20). Therefore, further research would be appropriate in determining how battling ropes can benefit specific patient populations.

Limitations to this study are due to the small sample size with individuals who had limited familiarity with the use of the ropes. The possibility that the acute metabolic changes seen in this study may be attenuated with individuals who are more trained with the use of the ropes. However, the use of trained individuals may allow for the BRP to be more advanced, thus resulting in acute changes that equal or surpass the levels seen in this study. This study used 2 independent groups with a repeated-measures design. Future studies may consider the use of a repeated-measures design that will assess the changes in $\dot{V}O_2$, HR and RPE across all 4 conditions for each subject to eliminate the subjects as a source of variability. The subjects in this study were relatively young and healthy. Future studies should measure these metabolic changes due to a BRP in a clinical population of individuals with neuromusculoskeletal impairments, which result in their inability to use their lower extremities in a rhythmical fashion to conduct aerobic exercise. In addition, testing the efficacy of the BRP on a healthy and clinical population as an intervention to enhance cardiovascular fitness using a variety of work-to-rest ratios and swing techniques would be the next logical step.

PRACTICAL APPLICATIONS

The main finding of this study was that the metabolic responses elicited were significantly lower for both BRPs as compared to the respective bicycle and treadmill tests. However, it can be concluded that battling ropes, whether performed in a seated or standing position, demonstrates the ability to produce acute metabolic responses that may enhance aerobic capacity. Thus, both the sitting and standing BRPs can offer an alternative form of exercise to improve cardiovascular endurance for individuals who cannot stand or move their lower extremities in a rhythmic manner to conduct aerobic exercise. These findings indicate that individuals who are unable to perform aerobic exercise

on a treadmill or bike could still meet their recommended exercise guidelines at a vigorous intensity. This is especially meaningful for people with disabilities who require an alternative form of exercise to maintain their cardiovascular fitness. For the strength and conditioning professional, battling ropes may be a suitable and affordable option to promote aerobic fitness for individuals who cannot stand. The increase in HR and oxygen consumption produced by the battling ropes may be replicated in a clinical population of individuals with neuromuscular diseases who cannot stand. If the work bouts are kept relatively short (10–15 seconds) during the initial phases of the treatment protocol and the rest periods are at least double to triple the work bouts, muscular fatigue can be reduced. Future studies will need to determine how to manipulate the work-to-rest ratios to allow for a training stimulus to occur to produce cardiovascular adaptations over time.

Battling ropes are relatively accessible and available to the public at most sporting goods stores. They can be used in almost any open area and are portable. Progressions of difficulty level can be tailored to the individual based on the diameter or length of the rope, the type of swing used, increasing the swing rate, and altering the work-to-rest ratios.

Further research is needed to determine the effectiveness of the battling ropes on cardiovascular fitness of individuals, both with and without disabilities.

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